Tests on Dualities using Global Symmetries and 't Hooft Anomalies

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In a Quantum Field Theory, it is possible that there is an obstruction to gauging a global symmetry. Such an obstruction is known as a 't Hooft anomaly in the system. Through 't Hooft anomaly matching condition, it is known that the anomaly of a system does not depend on a particular energy scale. This gives us a test on the recently conjectured infra-red dualities in 2+1 dimensions. In this paper, we give details of this philosophy and check it explicitly for the duality involving symplectic gauge groups.

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I. INTRODUCTION

Duality in Quantum Field Theory typically means that a particular system can be described from more than one point of view. Loosely speaking, they fall into two categories: ultra-violet dualities and infra-red dualities. Ultra-violet dualities are those for which the system has more than one description at any energy scale. Some examples of these are Ising model, and $SL(2,\mathbb{Z})$ duality of $\mathcal{N}=4$ supersymmetric Yang-Mills in four dimensions.

Infra-red dualities, on the other hand, are between two Quantum Field Theories that describe different Physics in ultra-violet, but flow to the same infra-red theory under renormalization. A well-known example of this kind of dualities is particle-vortex duality [1, 2] where two theories that describe different Physics in ultraviolet - the XY model and the Abelian Higgs model flow to the same IR CFT, the so-called $O(2)$ Wilson Fisher fixed point. This duality has been checked through lattice simulations [3, 4].

In the recent years, several infra-red dualities have been conjectured to exist in 2+1 dimensions. These involve various gauge groups, Chern-Simons levels and matter content in the form of scalars and fermions in the fundamental representation of the gauge groups. These dualities were first discovered through large N calculations for the case of unitary and special unitary gauge groups but have been since extended to include finite values of N and other simple Lie groups (see section II for more details) [5, 6].

In the absence of matter, these dualities are known to have been rigorously proven, and are known as level-rank dualities of Chern-Simons theories. Also known are the counter-parts of these dualities in supersymmetric theories where they are known as mirror symmetry in $\mathcal{N}=2$ and $\mathcal{N}=4$ theories in 2+1 dimensions. In presence of supersymmetry, several quantities can be computed in the two theories that are conjectured to be dual and shown to match, thus giving us more faith in the conjectures. There are not a lot of protected quantities, however, that one can exactly compute once the crutch of supersymmetry goes away.

One such test can, however, be made based on 't Hooft anomaly matching condition. In this paper, we review this recent attempt, first explained by the authors of [7]. A 't Hooft anomaly is as an obstruction to gauging a global symmetry of a quantum system, and it is well-known that these anomalies are independent of the energy scale at which they are computed. This gives us an ability to compare global symmetries and anomalies at weak coupling, and claim that they match at strong coupling.

Traditionally, 't Hooft anomalies in Quantum Field theories are known to exist in $SU(N)$ gauge theories in the presence of fermions, where they appear as inability to introduce a gauge-invariant regulator when we add background gauge fields for the global symmetry. Recently, it has been shown that these anomalies can also exist in systems that do not involve fermions - some times, in systems as simple as a particle on a circle or a Quantum Mechanical Gauged linear sigma model in presence of a Chern-Simons-like term. [7, 8] These examples shine light on all the different aspects of a system with anomaly, so we review them here.

This paper is organized as follows. In section II, we present in a more mathematical language the dualities that have been conjectured to exist in 2+1 dimensions. In section III, we describe 't Hooft anomaly in the quantum-mechanical system of a particle on a circle. This example is easy enough to solve, so we analyze it both in

Lagrangian and Hamiltonian formalism. In section IV, we describe 't Hooft anomaly in a quantum-mechanical gauged-linear sigma model with Chern-Simons-like term. This example exhibits all the important features that will appear again in the process of making checks on dualities. In section V, we discuss how this procedure gives check on dualities involving symplectic gauge groups. We also quote results for dualities involving other gauge groups.

II. DUALITIES IN 2+1 DIMENSIONS

We write here the conjectured dualities in 2+1 dimensions. All scalars and fermions transform in the fundamental representation of gauge groups, and all theories with scalars come with quartic potentials.

$$U(N)_{k,k\pm N} \text{ with } N_f \text{ scalars} \longleftrightarrow U(k)_{-N+\frac{N_f}{2},-N\mp k+\frac{N_f}{2}} \text{ with } N_f \text{ fermions} \quad (1)$$

$$SU(N)_k \text{ with } N_f \text{ scalars} \longleftrightarrow U(k)_{-N+\frac{N_f}{2}} \text{ with } N_f \text{ fermions} \quad (2)$$

$$U(N)_k \text{ with } N_f \text{ scalars} \longleftrightarrow SU(k)_{-N+\frac{N_f}{2}} \text{ with } N_f \text{ fermions} \quad (3)$$

The $\longleftrightarrow$ notation means that the theories on two sides are not dual each other in UV, but they describe the same Physics in IR. They are valid under the condition $N_f \leq N$. The unitary group $U(N)$ has two Chern-Simons levels coming from the $SU(N)$ and $U(1)$ parts: $U(N)_{k,k\pm N} \equiv \left(SU(N)_k \times U(1)_{N(k\pm N)}\right)/\mathbb{Z}_N$. When only one Chern-Simons level is mentioned it is understood that $U(N)_k \equiv U(N)_{k,k}$.

There also exist dualities for orthogonal and symplectic gauge groups:

$$SO(N)_k \text{ with } N_f \text{ real scalars} \longleftrightarrow SO(k)_{-N+\frac{N_f}{2}} \text{ with } N_f \text{ fermions} \quad (4)$$

$$USp(2N)_k \text{ with } N_f \text{ scalars} \longleftrightarrow USp(2k)_{-N+\frac{N_f}{2}} \text{ with } N_f \text{ fermions} \quad (5)$$

We will discuss global symmetries of these theories and their anomalies in section V.

III. PARTICLE ON A CIRCLE: A QUANTUM MECHANICAL EXAMPLE

A. 't Hooft anomaly in Lagrangian Formulation

Consider the following Lagrangian for a particle moving on a circle:

$$\mathcal{L} = \frac{1}{2}\dot{q}^2 \quad (6)$$

where q is a coordinate on S^1. This system has an $O(2)$ symmetry at classical level: given by an SO(2) action: $q \to q+\alpha$, $\alpha \in S^1$ and a $\mathbb{Z}_2$ parity transformation: $q \to -q$. Classically, a linear term in $\dot{q}$ does not change the equation of motion. But in Quantum Mechanics, we may modify our Lagrangian by a θ-term to get a one-parameter family of quantum systems:

$$\mathcal{L} = \frac{1}{2}\dot{q}^2 + \frac{\theta}{2\pi}\dot{q} \quad (7)$$

Under $\theta \to \theta+2\pi$, e^{iS} remains unchanged. It follows that at $\theta \neq 0, \pi$, the system has an $SO(2)$ symmetry, while at $\theta = 0, \pi$, it has an $O(2)$ symmetry.

Now, let us couple the system to an $SO(2)$ gauge field A. This modifies the Lagrangian as:

$$\mathcal{L} = \frac{1}{2}(\dot{q}+A)^2 + \frac{\theta}{2\pi}(\dot{q}+A) + kA \quad (8)$$

The last term in this Lagrangian is a Chern-simons-like coupling. It is linear in A because the system is one-dimensional. The $SO(2)$ gauge symmetry acts as $q \to q+\alpha(t)$, $A \to A - \dot{\alpha}$. It follows that the Chern-Simons term is gauge-invariant only when k is an integer.

Now as we shift $\theta \to \theta+2\pi$, we must change $k \to k-1$ to leave e^{iS} invariant. It follows that while the system has $O(2)$ symmetry at $\theta = 0$ for $k = 0$, it must have $k = -\frac{1}{2}$ at $\theta = \pi$ for the $\mathbb{Z}_2$ symmetry to survive. This is problematic because $k = -\frac{1}{2}$ does not give a well-defined Chern-Simons term. Hence, we say that the system has *'t Hooft anomaly* at $\theta = \pi$.

There are three different resolutions when a system exhibits 't Hooft anomaly. In the present example, these are:

1. Only a subgroup of the full global symmetry; in this case, $SO(2) \subset O(2)$ can be gauged.

2. We can set $A = 2B$ and couple the system to gauge field B instead of A. The gauge transformation acts on B as $B \to B - \frac{1}{2}\dot{\alpha}(t)$. Now α can go from 0 to 4π, which is equivalent to going to the double cover of $O(2)$. We call this group Pin(2).

3. We can attach a two-dimensional bulk M to the one-dimensional manifold X. Then the half-level Chern-Simons term may be written as a gauge-invariant action over the bulk.

$$\int_X \frac{1}{2}A = \int_M \frac{1}{2}F \quad (9)$$

where $\partial M = X$. Such a system is gauge invariant under the full global symmetry group $O(2)$. The bulk action by itself is not invariant under $\mathbb{Z}_2$ symmetry, but when attached to the boundary X, the full system has symmetry group $O(2)$. Such a phenomenon is commonly known as *anomaly inflow mechanism.*

It is worth noting that the action (9) depends on the properties of the bulk M and how we extend $SO(2)$-bundle over X to M. In particular, if we replace the bulk M with a different bulk M', the path integral gets a contribution of $e^{\frac{i}{2}\int_{\tilde{M}} F}$ where $\tilde{M}$ is the closed two-dimensional manifold made by gluing M and orientation reversal of M'. This is because

$$\int_M F = \int_{M'} F + \int_{\bar{M}} F \tag{10}$$

For any $U(1)$ gauge field, it is true that $\int_{\tilde{M}} F \in 2\pi\mathbb{Z}$. This implies that $e^{\frac{i}{2}\int_{\tilde{M}} F} = \pm 1$. We call this term *discrete θ-parameter* of the theory, as it encodes different ways in which the theory can be extended to the bulk.

B. 't Hooft anomaly in Hamiltonian Formulation

It is easy to find energy eigenstates and eigenvalues of the system described by (7).

$$E_n = \left(n - \frac{\theta}{2\pi}\right)^2 \tag{11}$$

for $n \in \mathbb{Z}_{\geq 0}$. For $\theta = 0$, there is a unique ground state corresponding to $n = 0$. At $\theta = \pi$, the ground state is degenerate.

Let the generator of $q \to q+\alpha$ be $\mathcal{O}_\alpha$ and the generator of $q \to -q$ be $\mathcal{O}_P$. These generators satisfy $\mathcal{O}_C\mathcal{O}_\alpha\mathcal{O}_C = \mathcal{O}_{-\alpha}$ Then at $\theta = \pi$, these operators act on the Hilbert space as:

$$\mathcal{O}_\alpha |n\rangle = e^{i\alpha n} |n\rangle \tag{12}$$

$$\mathcal{O}_P |n\rangle = |-n+1\rangle \tag{13}$$

It follows that $O(2)$ acts projectively on the Hilbert space. For a generic state $|n\rangle$, $\mathcal{O}_C\mathcal{O}_\alpha\mathcal{O}_C |n\rangle = e^{i\alpha}\mathcal{O}_{-\alpha} |n\rangle$. This is in contrast to $\theta = 0$ case, where, as it can be checked, the Hilbert space forms an ordinary representation of $O(2)$.

This gives an example of a well-known fact about 't Hooft anomalies: if a global symmetry of a system has a 't Hooft anomaly, the symmetry group acts projectively on the Hilbert space, and hence the ground state of the system must be degenerate.

IV. QUANTUM MECHANICS EXAMPLE II: GAUGED LINEAR SIGMA MODEL WITH 'CHERN-SIMONS' TERM

Consider the Lagrangian of two scalar fields charged under a $U(1)$ gauge field with potential and a Chern-Simons-like term.

$$\mathcal{L} = |D_b\phi^i|^2 - \frac{1}{4}\left(|\phi^1|^2 + |\phi^2|^2 - r\right)^2 + kb \tag{14}$$

where $D_b = \partial_t + ib$. As in the previous section, gauge invariance requires that $k \in \mathbb{Z}$. Naively, this Lagrangian has an $SU(2)$ global symmetry given by

$$\begin{pmatrix}\phi^1\\ \phi^2\end{pmatrix} \to \begin{pmatrix}a & b\\ c & d\end{pmatrix}\begin{pmatrix}\phi^1\\ \phi^2\end{pmatrix} \tag{15}$$

where $ad - bc = 1$ and $A^\dagger = A^{-1}$ for $A = \begin{pmatrix}a & b\\ c & d\end{pmatrix}$. But it is important to note that a $\mathbb{Z}_2$ subgroup of $SU(2)$ generated by $\begin{pmatrix}-1 & 0\\ 0 & -1\end{pmatrix}$ is also a subgroup of $U(1)$ - the gauge group of the theory. This implies that this subgroup will not act on gauge-invariant operators of the theory. As Physics is described purely by the correlation functions of gauge-invariant operators, we should really discuss global symmetry that acts faithfully on such operators. In this case, this group is $SU(2)/\mathbb{Z}_2 \equiv SO(3)$ instead of $SU(2)$.

Now we couple the system to gauge fields of the global symmetry. These gauge fields are valued in the Lie algebra $u(1) \oplus so(3)$. This does not uniquely identify the symmetry group. To identify the symmetry group, we consider action of $U(1) \times SU(2)$ on the scalar fields:

$$\begin{pmatrix}\phi^1\\ \phi^2\end{pmatrix} \to e^{i\theta}\begin{pmatrix}a & b\\ c & d\end{pmatrix}\begin{pmatrix}\phi^1\\ \phi^2\end{pmatrix} \tag{16}$$

Under this action, the element $\left(-1, \begin{pmatrix}-1 & 0\\ 0 & -1\end{pmatrix}\right) \in U(1) \times SU(2)$ acts as identity on the scalar fields. The subgroup of $U(1) \times SU(2)$ generated by this element is the largest subgroup that acts trivially on the scalar fields. This implies that the correct symmetry group is $\frac{U(1)\times SU(2)}{\mathbb{Z}_2} \equiv U(2)$.

This implies that we should couple the system to $U(2)$ gauge fields. In terms of $U(2)$ gauge field β, the Chern-Simons-like term is $kb = \frac{k}{2}\mathrm{Tr}\beta$. This is problematic for odd values of k as gauge invariance under $U(2)$ action implies that the co-efficient of $\mathrm{Tr}\beta$ should be an integer.

This leads to the conclusion that there exists a 't Hooft anomaly for symmetry group $U(2)$ for odd values of k. As discussed in the case of particle on a circle, one of the resolutions of 't Hooft anomaly is to attach a bulk and extend background gauge fields to the bulk. In this procedure, we should be careful that we do not extend $U(1)$ gauge fields to the bulk, as these are the fields that we integrate over in path-integral formalism, and the quantum dynamics of our system must remain one-dimensional.

As we attach a bulk M to the world-line X, the Chern-Simons-like term $\frac{k}{2}\int_X \mathrm{Tr}\beta$ can be written as $\frac{k}{2}\int_M \mathrm{Tr}F_\beta$. The latter is a gauge-invariant term, but as in the previous section, the price that we have to pay is that the choice of M is not unique. In particular, if we replace M by M', we get an extra factor of $e^{i\frac{k}{2}\int_{\tilde{M}} \mathrm{Tr}F_\beta}$ where $\tilde{M}$ is the closed two-manifold obtained by gluing M and $\bar{M}'$ (bar on top of M' in $\bar{M}'$ shows that it has orientation opposite to M'.

We can write $e^{i\frac{k}{2}\int_{\tilde{M}}\mathrm{Tr}F_\beta}$ as $e^{ik\int_{\tilde{M}}db}$. For a $U(1)$ gauge field, it is well-known that the fluxes are quantized as $\int_{\tilde{M}} db \in 2\pi\mathbb{Z}$. This may lead us to conclude that the factor $e^{i\frac{k}{2}\int_{\tilde{M}}\mathrm{Tr}F_\beta}$ is always 1 and hence, there is no dependence on the particular bulk that we attach. There is a subtlety in this argument that we have ignored.

The matter fields couple to the group $U(2) \equiv \frac{U(1)\times SU(2)}{\mathbb{Z}_2}$. Due to the presence of $\mathbb{Z}_2$ quotient, there is an extra possibility that must be considered. We can consider a globally ill-defined $U(1)$ 'bundle' in the bulk (for which the co-cycle condition gives -1 and the field strength satisfies $\int_{\tilde{M}}\frac{db}{2\pi}\in\frac{1}{2}\mathbb{Z}$) and an $SO(3)$ bundle that does not lift to $SU(2)$ bundle. Such an $SO(3)$ bundle will have non-trivial second Stiefel-Whitney class $w_2(SO(3))$ and hence, the co-cycle condition -1. Due to the $\mathbb{Z}_2$ quotient, $(-1,-1)\sim(1,1)$ and the two obstructions cancel to give a well-defined $U(2)$ bundle in the bulk.

This implies that $e^{i\frac{k}{2}\int_{\tilde{M}}\mathrm{Tr}F_\beta} = e^{ik\int_{\tilde{M}}db} = e^{ik\pi\int_{\tilde{M}}w_2(SO(3))} = \pm 1$. The dependence of the extension of $U(1)$ bundle to the bulk that we attach depends on the factor $e^{ik\pi\int_{\tilde{M}}w_2(SO(3))}$ that involves background gauge fields only and not the dynamical field b. We call this factor *discrete θ-parameter* as before.

V. CHECKS ON DUALITIES

A. Dualities involving symplectic groups

Consider a $USp(2N)$ gauge theory with N_f scalars in fundamental representation. Let us specialize to the case of $N = N_f = 1$. Since $USp(2)\equiv SU(2)$ this is the case of $SU(2)$ gauge theory with a scalar in fundamental representation $\mathbb{C}^2$. The isometry group of $\mathbb{C}^2\simeq\mathbb{R}^4$ is $SO(4)$. The Lie algebra of $SO(4)$ has a decomposition $su(2)\oplus su(2)$. As we gauge one copy of $SU(2)$ — let's call it $SU(2)_{\mathrm{dyn}}$, we naively expect the global symmetry of the theory to be the other copy of $SU(2)$ — we call it $SU(2)_{\mathrm{global}}$. However, the center of $SU(2)_{\mathrm{global}}$ is the same as the center of $SU(2)_{\mathrm{dyn}}$, and the global symmetry that acts on gauge-invariant operators is $SU(2)_{\mathrm{global}}/\mathbb{Z}_2\simeq SO(3)$. This implies that as we couple the system to background gauge fields, the matter fields couple to $\frac{SU(2)\times SU(2)}{\mathbb{Z}_2}$.

When the background gauge fields are ordinary $SU(2)$ gauge fields, the dynamical gauge fields are also $SU(2)$ gauge fields. But due to the $\mathbb{Z}_2$ quotient, we are also allowed the possibility of background $SO(3)$ gauge fields. In this case, the dynamical fields will also be $SO(3)$ gauge fields.

Now consider the case of generic N and N_f. In this case, since the fundamental representation of $USp(N)$ is pseudo-real, the global symmetry group is $USp(2N_f)$. But once again, the $\mathbb{Z}_2$ subgroup of $USp(2N_f)$ is gauged so the global symmetry group that acts faithfully on gauge-invariant operators is $USp(2N_f)/\mathbb{Z}_2$ and the matter fields couple to $\frac{USp(2N)\times USp(2N_f)}{\mathbb{Z}_2}$.

We can detect anomalies in this theory by imposing the condition that Chern-Simons levels must be properly quantized. In presence of Chern-Simons terms, the matter fields couple to $\frac{USp(2N)_k\times USp(2N_f)_L}{\mathbb{Z}_2}$ where $K\in\mathbb{Z}$. The quantization conditions are $Nk+N_fL\in 2\mathbb{Z}$ from taking the quotient and $L\in\mathbb{Z}$ from the integrality condition for $USp(2N_f)$ Chern-Simons levels.

Let us now look at the dual theory: $USp(2k)_{-N+\frac{N_f}{2}}$ with N_f fermions in fundamental representation. This theory also has a naive global symmetry group of $USp(2N_f)$ but the symmetry group that acts faithfully on gauge-invariant operators is $USp(2N_f)/\mathbb{Z}_2$. So we couple the matter fields to $\frac{USp(2k)_{-N+\frac{N_f}{2}}\times USp(2N_f)_{L+\frac{k}{2}}}{\mathbb{Z}_2}$.

Here we should make a comment about the factor $\frac{1}{2}$ in $\frac{N_f}{2}$. For a system of N_f fermions coupled to $USp(2k)_{-N+\frac{N_f}{2}}$, the bare Chern-Simons level is $-N+N_f$. The factor of half in $\frac{N_f}{2}$ is a convention based on the fact that integrating out N_f fermions gives Chern-Simons level of $-\frac{N_f}{2}$ in the effective action. However, the duality described above is valid before integrating out the fermionic fields.

This implies that the quantization condition for this case is also $Nk+N_fL\in 2\mathbb{Z}$ and $L\in\mathbb{Z}$. It follows that on both sides of the duality, there is an anomaly in the group to which matter fields couple, when $L\in\mathbb{Z}$ and $Nk+N_fL\notin 2\mathbb{Z}$.

As in the case of quantum-mechanical system, one can attach a bulk M here to the three-dimensional manifold X and quantify the anomaly in terms of discrete θ-parameters. It has been argued in [7] that the discrete θ-parameter associated with this anomaly is

$$e^{i\pi\int_{M_4}\frac{\mathcal{P}(w_2)}{2}} = \pm 1 \tag{17}$$

where w_2 is the second Stiefel-Whitney class of $USp(2N_f)/\mathbb{Z}_2$ bundle and $\mathcal{P}(w_2)$ is the Pontrjagin square of this class. The author does not know how to reproduce this result.

B. Another example: Duality involving unitary and special unitary groups

We quote here the results derived in [7] for one of the SU/U dualities. More details can be found in the paper cited here.

Each of the following two theories that are conjectured to be dual: $SU(N)_k$ with N_f scalars, and $U(k)_{-N+\frac{N_f}{2}}$ with N_f fermions has a global symmetry group $G_{\mathrm{faithful}} = U(N_f)/\mathbb{Z}_N$. Hence, in the former case, the matter fields couple to $\frac{SU(N)_k\times SU(N_f)_L\times U(1)_J}{\mathbb{Z}_N\times\mathbb{Z}_{N_f}}$. In the latter case, the matter fields couple to $\frac{SU(k)_{-N+\frac{N_f}{2}}\times SU(N_f)_{L+\frac{k}{2}}\times U(1)_{J+\frac{kN_f}{2}}}{\mathbb{Z}_k\times\mathbb{Z}_{N_f}}$. In both cases, the

quantization conditions are

$$L \in \mathbb{Z},\ J + kN \in k^2\mathbb{Z},$$
$$J - N_f L \in N_f^2\mathbb{Z},\ J \in kN_f\mathbb{Z} \tag{18}$$

VI. CONCLUSION

There are certain interesting topics that have not been discussed in this paper. The first one is the case of emergent symmetries. We have assumed throughout that for both sides of the duality, $G^{UV}_{\text{faithful}} = G^{IR}_{\text{faithful}}$, where $G^{UV/IR}_{\text{faithful}}$ are the global symmetry groups that act faithfully on gauge-invariant operators in ultra-violet and in infra-red. While it is true in most cases that these two groups match, exceptions exist, such as the duality between $U(1)_2$ with a scalar and $SU(2)_{-\frac{1}{2}}$ with a fermion. It follows from the arguments presented in the paper that the former theory has $O(2)$ global symmetry in UV while the latter theory has $SO(3)$ global symmetry. As they flow to IR, we conjecture that the $O(2)$ symmetry gets enhanced to $SO(3)$, and is only broken by relevant deformations in IR to $O(2)$.

Another interesting direction is to understand the comparison of 't Hooft anomalies in SO/SO duality given in equation 4. The authors of [7] have not discussed this case for reasons that we do not understand. This can lead to a project for more mathematically-minded physicists.

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